Impact of Vegetation Gradient and Land Cover Conditions on Soil Moisture Retrievals From Different Frequencies and Acquisition Times of AMSR2

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***Abstract*— Spaceborne remote sensing provides great potential for soil moisture (SM) retrieval and emerged as a significant data source for research in land surface dynamics and associated applications. This study compared the error characteristics of SM estimates retrieved from the Advanced Microwave Scanning Radiometer 2 (AMSR2) instrument on board National Aeronautics and Space Administration (NASA)'s Aqua satellite across different vegetation gradients and land cover conditions, at different overpass times and frequencies, to demonstrate their strengths and limitations. The results demonstrate that AMSR2 C-band products outperform AMSR2 X-band products over moderately and densely vegetated conditions due to lower attenuation by the vegetation canopy. Conversely, X-band products performed better than C-band products in barren lands possibly due to uneven sensing depth and microwave emissions from subsurface in C-band. The daytime products have a higher signal-to-noise ratio (SNR) in sparsely and moderately vegetated areas, whereas nighttime products have a higher SNR in densely vegetated areas. When these products are used selectively based on their error characteristics, the probability of obtaining SM with stronger signal than noise can be significantly improved (95%) at the expense of impaired spatial coverage (70% pixel loss).**

***Index Terms*— Advanced Microwave Scanning Radiometer 2 (AMSR2), error characteristics, microwave remote sensing, overpass time, soil moisture (SM), triple collocation analysis (TCA), vegetation.**

I. Introduction

SOIL moisture (SM) is a key state variable in comprehending and managing the water, energy, and carbon cycles [1], [2], [3], [4]. It regulates the water cycle by administering precipitation into infiltration and runoff; the energy cycle by bifurcating incoming radiation into sensible and latent heat fluxes; and the carbon cycle by modulating plant growth through carbon allocation, photosynthesis activity, and stomatal conductance [5], [6], [7], [8], [9], [10], [11]. Thus, accurate estimation of SM is extremely important in understanding and predicting various meteorological, hydrological, and agricultural phenomena [12], [13], [14], [15]. Advancements in remote sensing technology, together with the persistent demand for accurate, large-scale SM observations, have prompted the scientific community to establish regional and global SM products obtained from satellite remote sensing [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]. In recent years, a wide range of microwave-based SM products has been employed in such hydrometeorological applications as water resource management [28], [29], drought prediction [30], [31], [32], rainfall estimation [33], flood prediction [34], [35], [36], dust outbreak prediction [13], [37], and weather prediction [38], [39].

Currently, four passive microwave sensors, Soil Moisture Active and Passive (SMAP) radiometer, Microwave Imaging Radiometer with Aperture Synthesis onboard Soil Moisture and Ocean Salinity (SMOS), Microwave Radiation Imager onboard Fengyun-3B, and Advanced Microwave Scanning Radiometer 2 (AMSR2) onboard Global Change Observation Mission—Water Satellite 1, provide global-scale SM to the scientific community. Of these four passive microwave sensors, only AMSR2 provides SM at three different wavelengths: C1 (6.9 GHz), C2 (7.3 GHz), and X (10.7 GHz). SMAP and SMOS retrieve SM only from the L-band frequency, while Fengyun-3B retrieves SM only from the X-band frequency. All these products have been widely validated against in situ SM observations, revealing heterogenous performance from different frequency bands under different climate and land cover conditions [40], [41], [42], [43], [44]. Specifically, performance evaluations of AMSR2 SM products show that each band has its own strengths and weaknesses. Kaihotsu et al. [45] evaluated the AMSR2 level-2 SM products (10–36 GHz) over the Mongolian Plateau for a period of seven years, and they reported reasonable accuracy in terms of response patterns; however, they underestimated the in situ SM due to a measurement depth mismatch. Chen et al. [42] evaluated SMAP, SMOS, and AMSR2 SM against two observational networks on the Tibetan Plateau and reported contrasting performance from AMSR2 SM products (10–36 GHz) at two sites, suggesting a need for improvement in the products'

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algorithm. Kim et al. [46] comprehensively evaluated two global-scale AMSR2 products (X-band), the land parameter retrieval model (LPRM) and the Japanese Aerospace Exploration Agency (JAXA)'s algorithms, against 47 stations in a cosmic ray SM network. In terms of correlation, both products showed reasonable accuracy under low temperatures, in highly vegetated areas, and in highly wetted conditions. However, the JAXA product performed better than the LPRM product in drier conditions with regard to bias and root-mean-square error (RMSE). Wu et al. [47] evaluated AMSR2 SM products from different land cover types to generate a consistent spatial series of SM data. The results of their study showed that AMSR2 X-band SM had good accuracy in areas of low vegetation but it was less accurate in areas of dense vegetation. Similarly, several researchers evaluated the reliability of AMSR2 SM products for climate applications [31], [48], [49]. Although many studies have evaluated individual AMSR2 products against ground-based observations or have compared AMSR2 products with other microwave-based SM products, no study exist to date which intercompared AMSR2 products based on different wavelengths and overpass times, specifically on a global scale.

The use of different frequency bands for SM retrieval is associated with depth of measurement, impact of vegetation canopy, and surface roughness. Low-frequency bands (e.g., L-band) are assumed to be best suited for SM retrievals [16], [50]; however, these bands may produce poorer quality data due to radio frequency interference (RFI). On the other hand, SM retrievals from ascending and descending overpass times may differ in quality due to diurnal variation in underlying surface conditions. Most of the literature inclines toward the use of nighttime or early morning overpass products [51], [52], [53], [54] with the expectation of stable atmospheric conditions at those times [55]. Influence from Faraday rotation is also assumed to be minimal during nighttime overpasses [56]. However, research has shown that changes in daytime vegetation water content (VWC) influence SM retrievals from space [57], [58] because VWC attenuates microwave emissions from the soil. However, the vegetation canopy loses water in the daytime due to evapotranspiration, thus making it relatively transparent [59], while research has shown RFI to be most severe during the nighttime, making nighttime SM retrievals problematic [60], [61], [62]. All things considered, both ascending and descending overpass time products have their strengths and shortcomings. Their performance depends on the underlying surface conditions of the study area. Since AMSR2's overpass time is not optimized for SM retrieval from space [53], [63], examination of its error characteristics and the limitations of its products at different overpass times is important. This knowledge will enable users to utilize AMSR2's products with maximum accuracy by helping them identify its potential shortcomings and will assist in improving its algorithm [64].

Satellite-based SM products must be validated in order to ensure their quality; however, due to the lack of high-quality reference data at satellite footprints and global scale coverage, validation can be a difficult task [65]. Traditionally, satellite-based SM products have been validated against in situ SM observations. The main problem with this type of validation is the vast difference in spatial scales (25 km versus point scale) and uneven distribution in different land cover and vegetation classes. Furthermore, representative layer depth differs between in situ (5–10 cm) and satellite-based SM (0.5–2 cm). The primary goal of this study was to evaluate the error characteristics of AMSR2 products across different vegetation conditions and land cover types, so we employed triple collocation analysis (TCA) [66]. Recently developed evaluation techniques such as TCA give researchers the opportunity to assess the relative accuracy of satellite-based SM products without the use of in situ observations. Specifically, in this study, we evaluated the accuracy of ascending and descending overpass retrievals from AMSR2's different bands (C1—6.9 GHz, C2—7.3 GHz, and X—10.7 GHz) using TCA together with SM data from Advanced Scatterometer (ASCAT) and Global Land Data Assimilation System (GLDAS). We used Copernicus ASCAT soil water index (SWI) data which is based on the TU-Wien algorithm [12]. Interpretation of ascending and descending overpass SM products will not only enable us to merge the products for maximum accuracy, but it may also provide guidelines for using these products in various applications and decision-making processes for disaster identification, prevention, and mitigation. Moreover, both ascending and descending overpass SM products can be used systematically in applications such as daily temperature change analysis, agricultural yield estimation, and irrigation scheduling [63], [67]. The primary objective of this study is to intercompare AMSR2 SM products (i.e., X-band, C1-band, and C2-band) in terms of their error characteristics and to propose synergistic use of these products under different surface conditions in a manner which has not been discussed before. As the frequencies used by AMSR2, like those of other passive microwave sensors, have unique strengths and weaknesses in retrieving SM and while the microwave remote sensing community has been using X- and C-band AMSR2 products without proper direction, this information is necessary to maximize the use of AMSR2 SM data.

II. Datasets

A. Advanced Microwave Scanning Radiometer 2

After National Aeronautics and Space Administration (NASA)'s discontinuation of the Advanced Microwave Scanning Radiometer for the Earth Observation System (AMSR-E) on board the Aqua spacecraft, the JAXA launched the AMSR2 sensor on board the GCOM-W1 satellite in May 2012 [68], [69]. The primary objective of AMSR2 is to observe water cycle components such as precipitation, water vapor, and SM. AMSR2 is a passive microwave radiometer that scans the entire globe twice a day, every one to two days, with an ascending overpass time of 13:30 and a descending overpass time of 01:30 at a spatial resolution of $0.25^\circ \times 0.25^\circ$. It shares the same frequency range as AMSR-E with the exception of an additional C-band to counter RFI problems [60], [70], [71]. SM is retrieved from three frequency bands: 6.9 (C1), 7.3 (C2), and 10.7 GHz (X). Two types of AMSR2 SM products are

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available based on their retrieval algorithms: JAXA [72], [73] and LPRM [74]. The JAXA algorithm is used to retrieve SM products only in the X-band, whereas LPRM is used to retrieve SM and vegetation optical depth (VOD) simultaneously in both the C- and X-bands. Since the LPRM algorithm retrieves all SM products in all bands, we used only LPRM SM and VOD products for this study.

B. Advanced Scatterometer

The ASCAT sensors onboard the Meteorological Operational Platform (MetOP) satellites are active sensors operating in the C-band (5.3 GHz). Originally, three MetOP satellites carrying ASCAT sensors are being used to provide SM: MetOP-A, MetOP-B, and MetOP-C, which were launched in 2006, 2012, and 2018, respectively. However, currently, MetOP-B and MetOP-C are operating, whereas MetOP-A has been deorbiting since November 2021. ASCAT's local overpass times for its ascending and descending orbits are 21:30 and 09:30, respectively. Backscattered electromagnetic signals measured by two sets of three antennas are converted to SM using change detection algorithms developed by the Vienna University of Technology, Vienna, Austria [75]. ASCAT provides SM at 25- and 50-km spatial resolution and covers the entire globe every one to two days [12]. However, backscattering signals in the C-band are affected by surface roughness and vegetation influences the quality of ASCAT SM retrievals [76], [77]. TU Wien algorithm estimates ASCAT's SM in terms of degree of saturation ranging from 0% (dry land) to 100% (wet land). We used ascending/descending ASCAT SM data and matched the overpass time with AMSR2 overpass time. The degree of saturation is then converted to volumetric units by multiplying it with porosity values derived from the harmonized world soil database (HWSD) [78], [79].

C. VWC From SMAP

The NASA's SMAP is an L-band radiometer dedicated to measuring global SM [80]. VWC, an essential ancillary data for SM retrieval from SMAP, is derived from Moderate Resolution Imaging Sensor's normalized difference vegetation index (NDVI) climatology which illustrates the VWC of the above-ground canopy biomass [81], [82]. VWC is an important ecological indicator that can potentially measure the microwave attenuation of SM as it passes through a vegetation canopy [83]; thus, it can indicate vegetation conditions. SMAP can effectively estimate VWC because of its advantage in mitigating RFI, its insensitivity to cloud liquid water, and its deep penetration into the vegetation canopy [84]. Thus, in this study, SMAP VWC was used to evaluate AMSR2's ascending and descending overpass products in terms of vegetation fraction. First, barren land was masked out (only for defining vegetation classes based on VWC) using the International Geosphere Biosphere Program (IGBP) land cover classification. Next, based on the empirical distribution function, we defined sparsely, moderately, and densely vegetated regions. Sparsely vegetated regions were defined as those in which the VWC was greater than 0.1 $kg{\cdot}m^{-2}$ but less than 33% (0.73 $kg{\cdot}m^{-2}$), moderately vegetated regions were defined as those in which the VWC was greater than 33% (0.73 $kg{\cdot}m^{-2}$) but less than 66% (4.35 $kg{\cdot}m^{-2}$), and densely vegetated regions were defined as those in which the VWC was greater than 66% (4.35 $kg{\cdot}m^{-2}$). Fig. 1 shows the average VWC values during the study period (from 2015 to 2019).

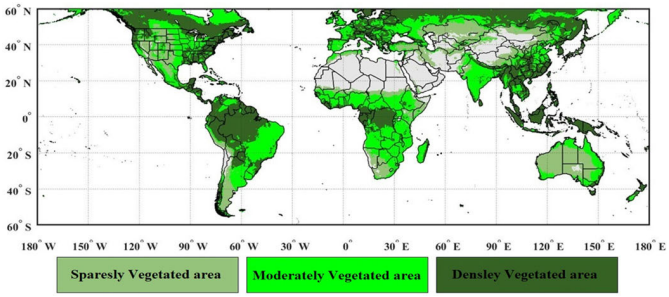


Fig. 1. Global maps of mean VWC from SMAP L-band for the study period (2015–2019). (For interpretation of the colors used in this figure's legend, the reader is referred to the web version of this article.)

D. Global Land Data Assimilation System

The GLDAS [85] is a global reanalysis dataset developed by NASA. It produces optimal land surface states and fluxes by combining several satellite- and ground-based observations using complex land surface models and data assimilation techniques in the land information system framework [86]. In this study, GLDAS version 2.1 Noah SM data from top soil layer (0–10 cm) was used to perform TCA along with AMSR2 and ASCAT products. The Noah model provides atmospheric and land surface states on a regular grid of 0.25° × 0.25° spatial resolution and 3 hourly temporal resolution. As GLDAS SM products are based on the universal time coordination (UTC) system, the UTC-based GLDAS products are matched with the local overpass times of AMSR2.

E. IGBP Land Cover Classification

The IGBP provides global land cover classification at 1-km spatial resolution to satisfy the need of high-resolution land cover classification for various projects related to climate, carbon, water, and energy cycles [87]. To produce the IGBP land cover dataset, both high-resolution satellite imagery and ground observations are used [88]. The IGBP Land Cover Working Group depicts 17 land cover classes. Greenness classes are identified based on the ten-day NDVI data from the Advanced Very High-Resolution Radiometer (AVHRR) and are further separated using unsupervised classification clustering. AVHRR Channel 2 is used to identify the water bodies class, while low NDVI 12-month composite values are used as signatures for snow and ice, barren land, and sparsely vegetated pixels. The urban and built-up classes are separated by using "urban" layer information from the Digital Chart of the World. For detailed methodology about the classification system, readers are referred to [89]. Fig. 2 shows the global map of IGBP land cover classification used in this study.

All the satellite-based datasets were resampled using the nearest neighbor distance algorithm to establish a uniform georeferenced 0.25° grid, which is the same grid used in the GLDAS datasets [41], [90].

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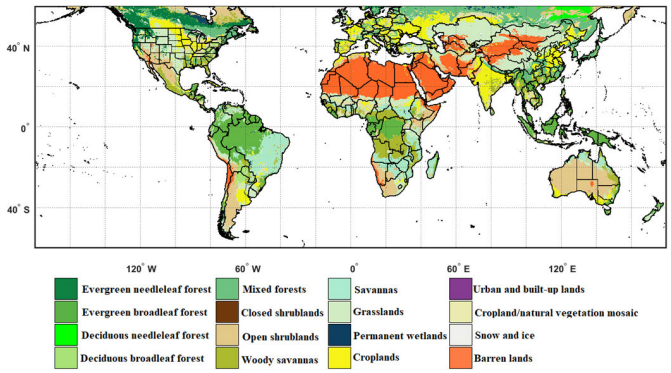


Fig. 2. Global maps of IGBP land cover classification. (For interpretation of the colors in this figure's legend, the reader is referred to the web version of this article.)

III. Methodology

A. Triple Collocation Analysis

The primary goal of this study was to evaluate the error characteristics of AMSR2 products in different vegetation conditions and land cover types rather than to directly compare AMSR2 products with ground observations. In general, very few SM networks have more than ten stations within 25-km pixel of AMSR2 data.

Stoffelen [91] introduced the concept of TCA to compare the mutually independent errors of three collocated observations of a geophysical variable by using RMSE matrices. Scipal et al. [92] were the first to use this concept to evaluate the error variance of large-scale SM datasets acquired from different sources. Since then, TCA has been used extensively to evaluate the error characteristics of SM datasets from different sources such as ground observations, active and passive microwave remote sensing, and reanalysis products [41], [66], [93], [94]. The basic assumption of TCA is that collocated measurements X_i are related to the underlying true state T via an affine error model

$$x_i = \alpha_i + \beta_i T + \varepsilon_i \tag{1}$$

where T is the true SM signal, i refers to different SM measurements from, for instance, active SM products, passive SM products, and model-based SM products. α_i and β_i are the calibration constants, and ε_i denotes random errors in the measurement system. SM RMSEs can only be derived by assuming that mutual independence within the triplet is fulfilled. Depth mismatch of different SM products can violate the presumption of TCA; however, this will have only a minor impact on the TC-based correlation analysis [95].

Rescaling the SM products enables the comparison of products with different units, sensitivity, and measurement depth, thus avoiding the dependence of scaled error variance patterns on the spatial climatology of the chosen reference product [66]. Thus, before calculating the error variance ($\sigma_{\varepsilon_i}^2$), we removed climatology from the SM data [96] and calculated the error variance as follows:

$$\sigma_{\varepsilon_x}^2 = \sigma_x^2 - \frac{\sigma_{xy}\sigma_{xz}}{\sigma_{yz}}$$
$$\sigma_{\varepsilon_y}^2 = \sigma_y^2 - \frac{\sigma_{yx}\sigma_{yz}}{\sigma_{xz}}$$
$$\sigma_{\varepsilon_z}^2 = \sigma_z^2 - \frac{\sigma_{zx}\sigma_{zy}}{\sigma_{xy}} \tag{2}$$

where σ_i^2 ($i \in [x, y, z]$) is the variance of the data and σ_{ij} is the covariance of the data between two SM products ($i, j \in [x, y, z]$ and $i \neq j$). To quantify the relative error variance of AMSR2 SM products, the fractional mean-square-error (fMSE) metric was calculated [95]. This fMSE metric is useful because a value of fMSE lower than 0.5 indicates that the SM signal is stronger than its noise. The fMSE variance was calculated in the following equation:

$$\text{fMSE}_i = \frac{\sigma_{\varepsilon_i}^2}{\sigma_i^2} = \frac{\sigma_{\varepsilon_i}^2}{a_i^2\sigma_\theta^2 + \sigma_{\varepsilon_i}^2} = \frac{1}{\text{SNR} + 1} \tag{3}$$

where $i, j \in [x, y, z]$ and signal-to-noise ratio (SNR) is $a_i^2(\sigma_\theta^2/\sigma_{\varepsilon_i}^2)$. x is an AMSR2 SM product (i.e., C1-, C2-, and X-band), and y and z are ASCAT and GLDAS SM data, respectively. Please refer to [66], [97], and [98] for further information. In this study, we focused only on the statistically stationary random error of satellite-based SM datasets. In a future study, we will investigate the random errors in AMSR2 SM products for different time periods (e.g., autumn, winter, spring, and summer) for each satellite-based SM product using datasets of longer time periods [99].

Additionally, the reliability of TC-based results was examined by comparing them with the conventional fMSE metric calculated as

$$\text{fMSE} = \frac{E\left[\left(\hat{\theta} - \theta\right)^2\right]}{E\left[\left(\hat{\theta} - E\left[\hat{\theta}\right]\right)^2\right]} \tag{4}$$

where $E[\cdot]$ computes the expectation values, $\hat{\theta}$ represents the SM from AMSR2 and ASCAT, and θ is the SM obtained from GLDAS. We used International Soil Moisture Network (ISMN) ground-based SM data, specifically, ISMN stations which measure SM at a depth less than or equal to 5 cm and provide data for the study period (2015–2019). A total of 1357 ground-based stations were tested. All datasets were quality controlled [100] and cross-screened to narrow the data to grid pixels for which all six AMSR2 datasets were readily available. To obtain robust statistical results, we excluded SM stations with fewer than 100 data points in the time frame of the corresponding grid pixels. After preprocessing, data from 85 sites were viable, the majority of which were located in moderately vegetated areas (61) and densely vegetated areas (16). Only eight stations in this study were in sparsely vegetated areas. We created boxplots if at least ten datasets were available.

Moreover, to remove the negative effect of RFI contamination and to obtain reliable conclusions, RFI contaminated pixels were masked out based on the study of [70].

IV. Results and Discussion

A. Spatial Distribution of Error Characteristics of AMSR2

Fig. 3 shows the spatial distribution of fMSE-based accuracy differences for all three SM products of AMSR2: C1-band

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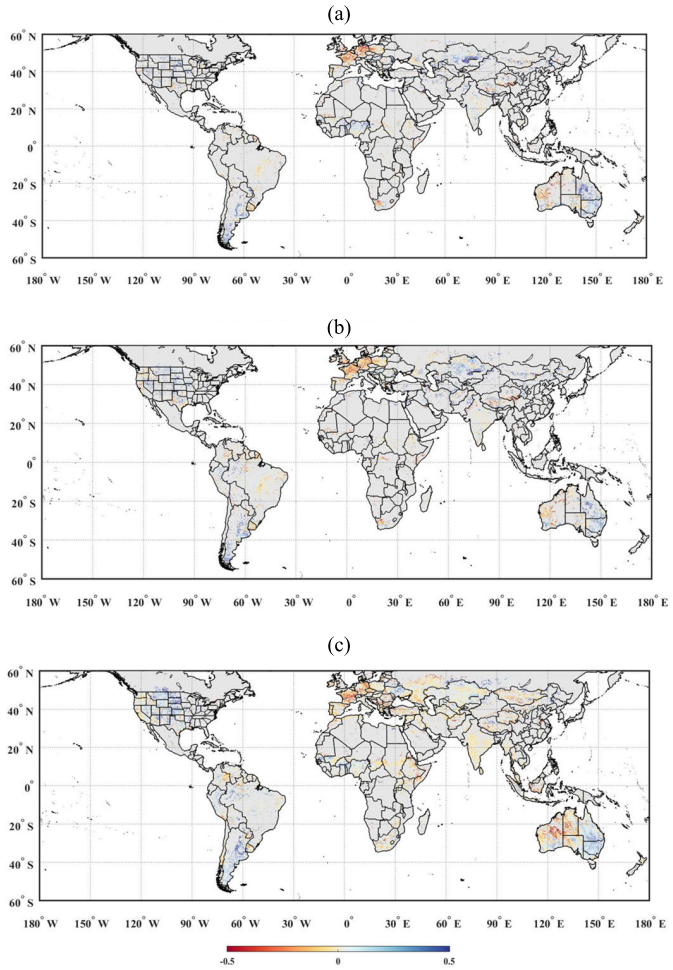


Fig. 3. Difference between fMSE values of AMSR2 daytime and nighttime SM products calculated as $fMSE_{nighttime} - fMSE_{daytime}$: (a) C1-band; (b) C2-band; and (c) X-band. The positive (negative) values indicate that the daytime (nighttime) products are less noisy. (For interpretation of the colors in this figure's legend, the reader is referred to the web version of this article.)

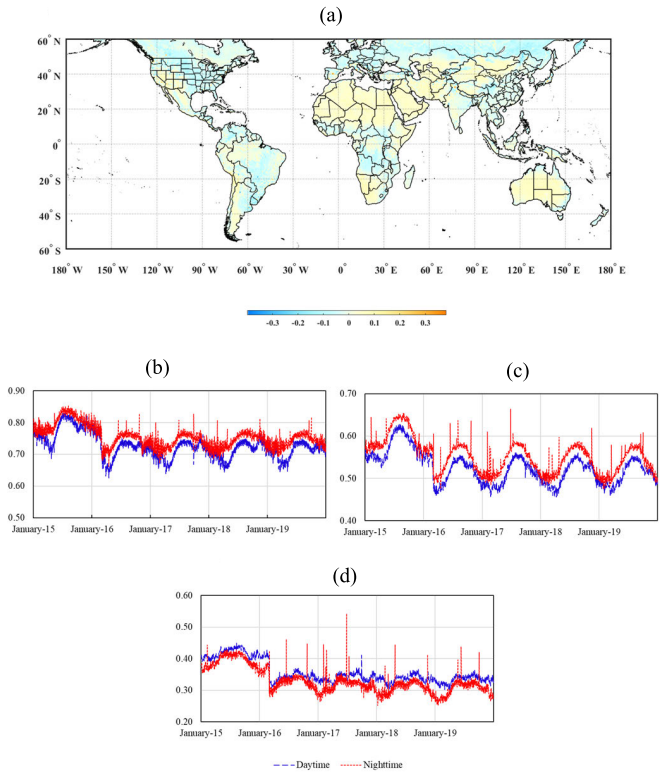


Fig. 4. Diurnal variation of daytime and nighttime VOD: (a) spatial distribution of daytime VOD minus nighttime VOD; (b) temporal variation of daytime and nighttime VOD in densely vegetated areas; (c) temporal variation of daytime and nighttime VOD in moderately vegetated areas; and (d) temporal variation of daytime and nighttime VOD in sparsely vegetated areas. (For interpretation of the colors in this figure's legend, the reader is referred to the web version of this article.)

[Fig. 3(a)], C2-band [Fig. 3(b)], and X-band [Fig. 3(c)]. The difference was calculated as $fMSE_{nighttime} - fMSE_{daytime}$, so the positive value indicates that the daytime product outperformed the nighttime product, and the negative value indicates that the nighttime product outperformed the daytime product. Values close to zero indicate that the products performed equally well.

The spatial distribution of fMSE shows that the difference between daytime and nighttime products was not very prominent in Asia and Europe (fMSE differences ranged from −0.2 to 0.2), whereas in Australia and North and South America, daytime products outperformed nighttime products. Furthermore, when the spatial distribution of fMSE (Fig. 3) was compared to the VWC map (Fig. 1), the daytime products outperformed the nighttime products in sparsely and moderately vegetated regions. This could be explained by the well-known fact that during the daytime, plants are transparent due to water loss via evapotranspiration, allowing more accurate retrieval of satellite-based SM. To validate this assumption, a sensitivity analysis with a diurnal variation of VOD on the performance of different wavelengths and overpass times was performed. Fig. 4(a) shows the spatial distribution of mean daytime VOD minus mean nighttime VOD and that the difference is mostly negative in vegetated regions, indicating that daytime VOD is lower than nighttime VOD during the study period. The time-series comparison of daytime VOD and nighttime VOD for 2015–2019 for three vegetation classes [Figs. 4(b)–6(d)] shows that daytime VOD is less than nighttime VOD in densely and moderately vegetated areas, where, in sparsely vegetated areas, daytime VOD is slightly higher than nighttime VOD. The sparsely vegetated areas mostly consist of open shrubs, which are more isohydric [81], and keep leaf water potential relatively steady under water stress conditions [101]. Various other researchers also reported higher average daytime VOD than nighttime VOD in different climate regions [81], [102], [103]. Another factor that contributes to better daytime SM products over nighttime products is dew formation and rainfall interception. As a result of dew formation and rainfall interception, the microwave emission from the soil is attenuated and its own emissions are added, possibly resulting in erroneous SM retrieval at nighttime [104], [105], [106]. Moreover, deep rooted plants in wet climate conditions follow the hydraulic lift phenomenon in which plants release water to the soil during the nighttime triggered by reduced root pressure to maintain equilibrium with soil matric potential [107], [108], [109].

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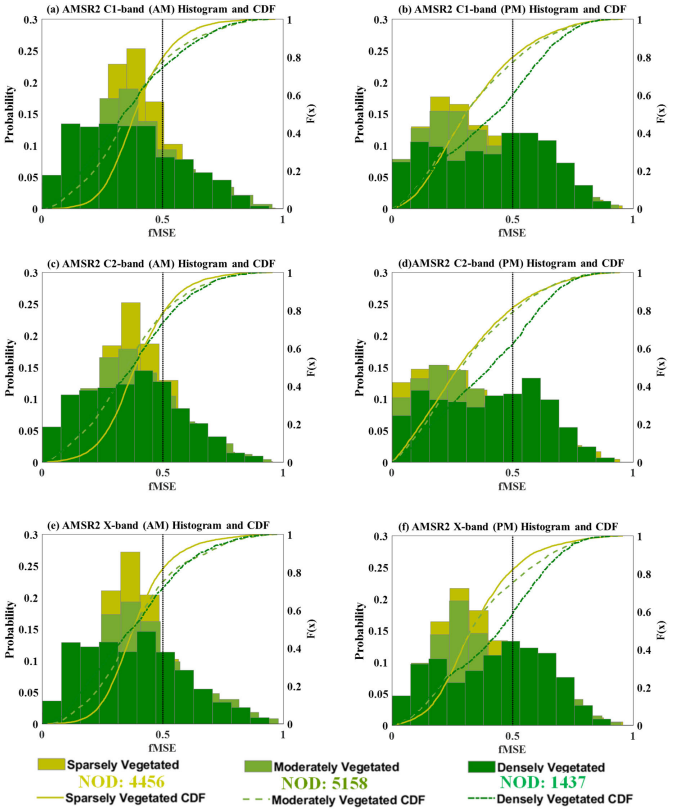


Fig. 5. Probability histogram and cumulative distribution function (CDF) of fMSE values over sparsely, moderately, and densely vegetated conditions for: (a) AMSR2 C1-band daytime; (b) AMSR2 C1-band nighttime; (c) AMSR2 C2-band daytime; (d) AMSR2 C2-band nighttime; (e) AMSR2 X-band daytime; and (f) AMSR2 X-band nighttime. (For interpretation of the colors in this figure's legend, the reader is referred to the web version of this article.)

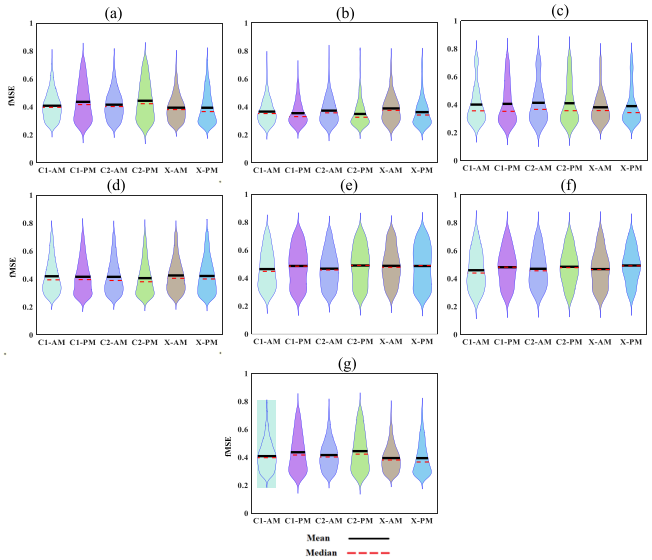


Fig. 6. Violin plots showing the error characteristics of AMSR2 products based on land cover classification: (a) open shrublands (number of data points (NOD): 3148); (b) woody savannas (NOD: 608); (c) savannas (NOD: 616); (d) grasslands (NOD: 5623); (e) croplands (NOD: 3308); (f) cropland/natural vegetation mosaics (NOD: 856); and (g) barren lands (NOD: 966). (For interpretation of the colors in this figure's legend, the reader is referred to the web version of this article.)

The results of this study are in line with [95] and [110], who reported that daytime products were more reliable than

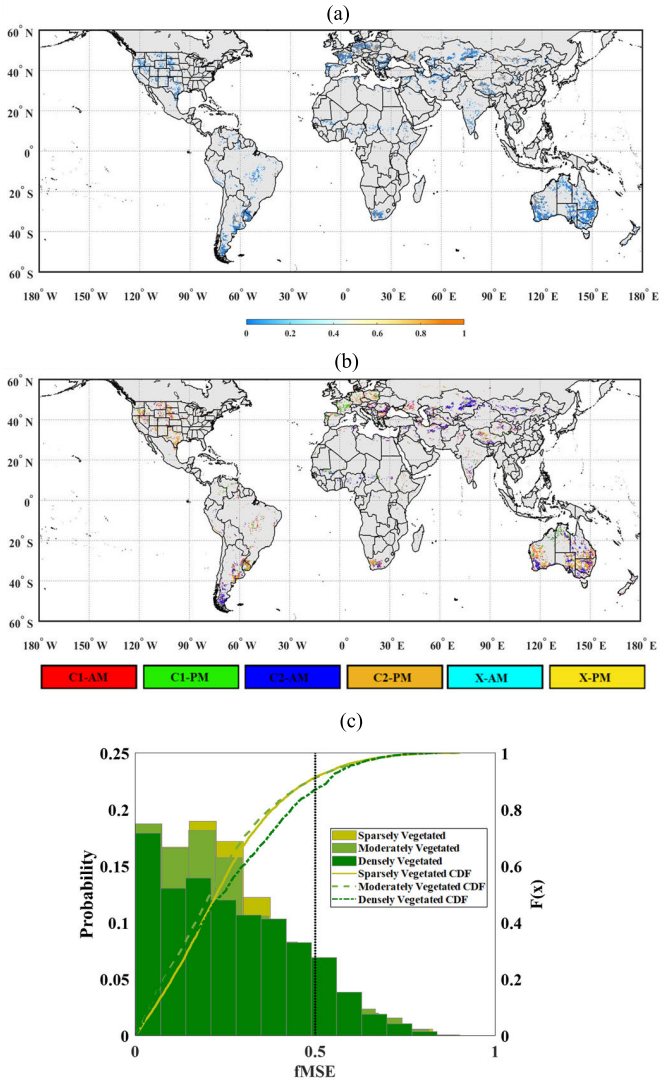


Fig. 7. (a) Global map of AMSR2 products used selectively based on fMSE values. (b) Global map of AMSR2 frequency bands and overpass time products after selective use. (c) Probability histogram and CDF of fMSE values for combined data over sparsely, moderately, and densely vegetated regions.

nighttime products. Similarly, Brocca et al. [111] reported that daytime overpasses produced better results over Europe for certain types of vegetation density; however, more in-depth studies are needed to assess this important finding. Furthermore, our findings contradict those of [63], who reported better performance of nighttime overpass products in sparsely vegetated areas with regard to the correlation metric (western USA). In this study, based on the fMSE metric, we found that daytime overpass products performed better than nighttime overpass products in sparsely and moderately vegetated areas.

To check the possible seasonality of AMSR2 SM products, their performance was evaluated over three latitude belts as: high-mid-latitude cells (30°N–60°N), Hadley cells (30°S–30°N), and low-mid-latitude cells (60°S–30°S). The histogram and cumulative distribution of median fMSE (Fig. 8 and Table II) show that all three AMSR2 bands (C1, C2, and X) showed relatively poor performance over

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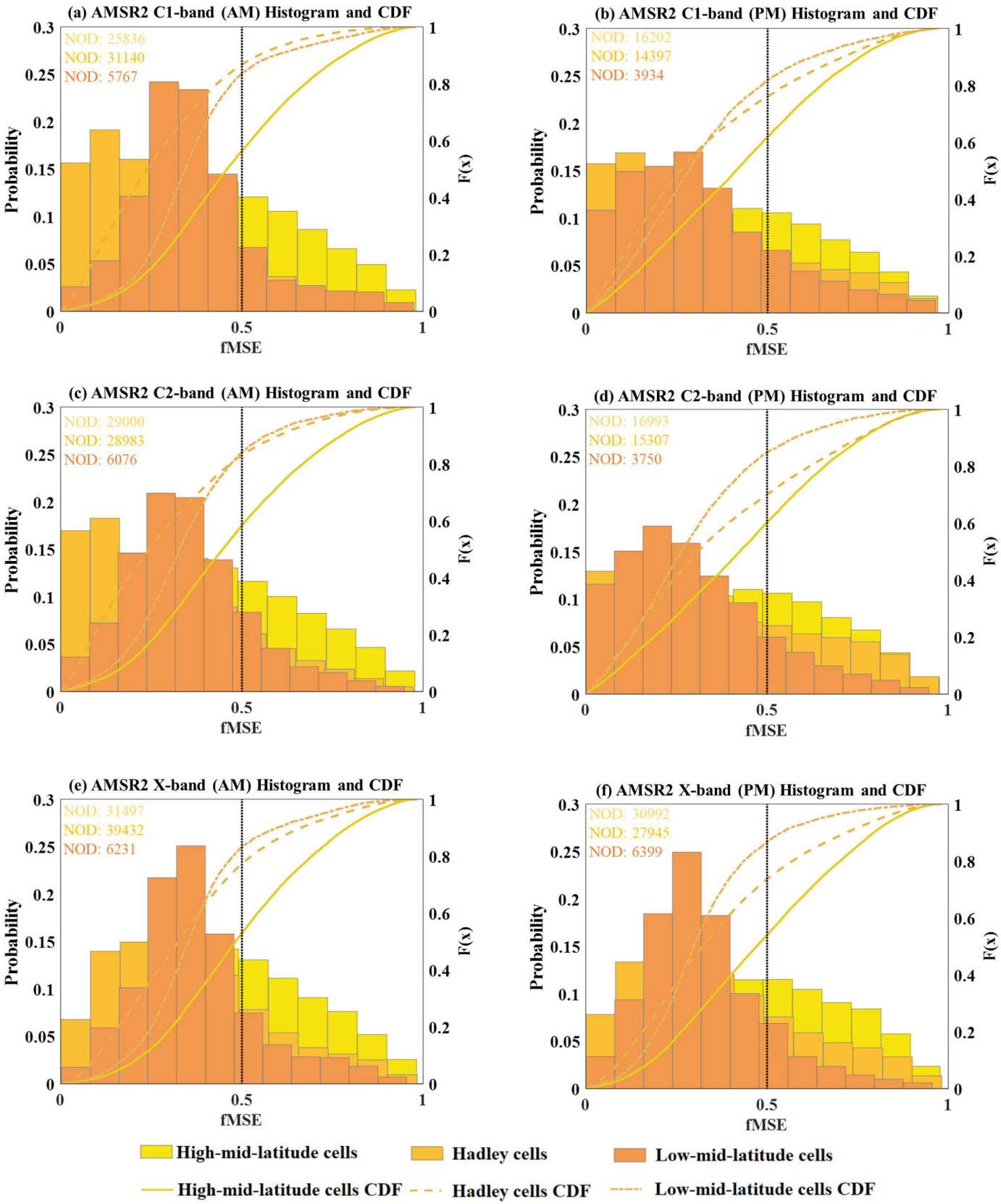


Fig. 8. Probability histogram and CDF of fMSE values for (a) AMSR2 C1-band daytime, (b) AMSR2 C1-band nighttime, (c) AMSR2 C2-band daytime, (d) AMSR2 C2-band nighttime, (e) AMSR2 X-band daytime, and (f) AMSR2 X-band nighttime over High-Mid-latitude cells, Hadley cells, and Low-Mid-latitude cells. (For interpretation of the colors in this figure's legend, the reader is referred to the web version of this article).

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TABLE I
$F_{fMSE}(0.5)$ VALUES OVER SPARSE, MODERATE, AND DENSE VEGETATION CONDITIONS

Bands / Vegetation	Acquisition time	F_{fMSE} (0.5|Sparse)	F_{fMSE} (0.5|Moderate)	F_{fMSE} (0.5|Dense)
C1	NIGHTTIME	0.7969	0.7739	0.7437
	DAYTIME	0.7979	0.7721	0.5993
C2	NIGHTTIME	0.7899	0.7824	0.7373
	DAYTIME	0.8133	0.7909	0.6233
X	NIGHTTIME	0.8194	0.7509	0.7177
	DAYTIME	0.8221	0.7544	0.5891

TABLE II
$F_{fMSE}(0.5)$ VALUES OVER HIGH-MID-LATITUDE, HADLEY CELLS, AND LOW-MID-LATITUDE CELLS

Bands / Vegetation	Acquisition time	F_{fMSE} (0.5|High-mid latitude cells)	F_{fMSE} (0.5|Hadely cells)	F_{fMSE} (0.5|Low-mid latitude cells)
C1	Nighttime	0.5638	0.8673	0.8359
	Daytime	0.6187	0.7610	0.8181
C2	Nighttime	0.5850	0.8310	0.8416
	Daytime	0.6030	0.7010	0.8486
X	Nighttime	0.5289	0.7735	0.8318
	Daytime	0.5395	0.7368	0.8682

high-mid-latitude (F_{fMSE} ranges 0.5–0.6), whereas in Hadley cells and low-mid-latitude, all three SM products showed good performance (F_{fMSE} ranges 0.7–0.8). In terms of overpass time, the daytime products outperformed nighttime products over high-mid-latitude for all three SM products and X-band products in low-mid-latitudes, whereas nighttime products outperformed daytime product in Hadley cells for all three AMSR2 SM products and for C-band in low-mid-latitude cells. The error characteristics of AMSR2 based on different frequency bands and overpass time over varying vegetation conditions are discussed in detail in Sections IV-B–IV-E.

B. Error Characteristics Comparison of Different Frequency Bands

Fig. 5 shows the histogram and cumulative distribution of median fMSE with regard to different vegetative conditions for the three AMSR2 bands: C1 (row 1), C2 (row 2), and X (row 3), together with their respective overpass times (descending at left and ascending at right). In general, the empirical distributions showed a wide dispersion with a low mean, a positive skewness, and high kurtosis. The $F_{fMSE}(0.5)$ values of each plot indicate the probability of the SM data having stronger signal than noise [96]. Table I shows the results of $F_{fMSE}(0.5)$ for different products in various vegetation gradients.

In general, all six AMSR2 products (three bands and two overpasses) showed F_{fMSE} values greater than 0.6 across all vegetation types. Specifically, $F_{fMSE}(0.5|$ moderate) ranged from 0.7 to 0.8 for C-band products and 0.6–0.8 for X-band products. The three bands' SM products showed comparable performance in sparsely and moderately vegetated areas, whereas in densely vegetated areas, the C-band products showed relatively better performance than the X-band products. These findings depict that AMSR2 C-band products outperformed AMSR2 X-band product over moderately to densely vegetated regions, thus satisfying the theoretical concept that higher frequency microwaves can penetrate dense vegetation canopy [112].

Additionally, the results of TC-based fMSE with the conventional fMSE (Fig. 9) show that the TC-based fMSE accurately represents the conventional fMSE. However, the results may be misleading for the two reasons. First, there were few available ISMN datasets in the sparsely and densely vegetated areas. Second, even though some stations correspond to sparsely and densely vegetated pixels, they are not true representatives of these areas due to the heterogeneous nature of the satellite pixels. In densely vegetated areas, these stations are most often

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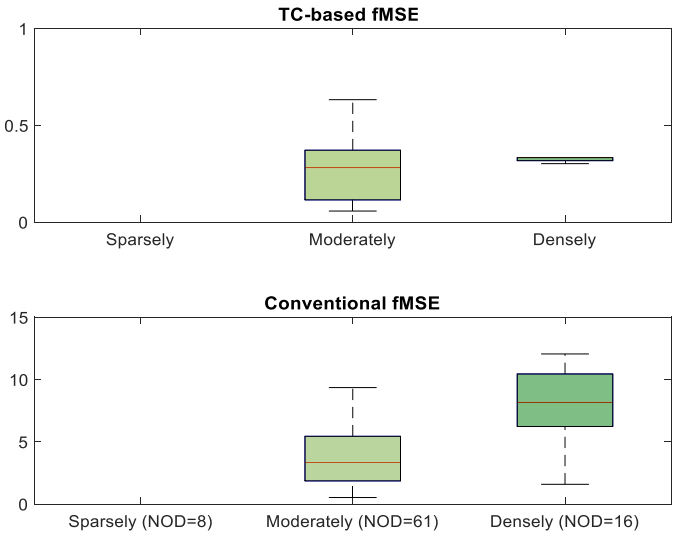


Fig. 9. Boxplot of the TCA-based fMSE and the conventional fMSE for sparsely, moderately, and densely vegetated areas. (NOD means number of data points used for boxplot).

located outside forests or in open spaces within tree stands; thus, they do not capture the SM variability caused by forest canopy micrometeorology.

C. Error Characteristic Comparison Based on Overpass Time

In a comparison of different overpass time products, daytime products outperformed nighttime products for sparsely ($0.1\ \text{kg}\cdot\text{m}^{-2} < \text{VWC} < 0.73\ \text{kg}\cdot\text{m}^{-2}$) and moderately ($0.73\ \text{kg}\cdot\text{m}^{-2} < \text{VWC} < 4.35\ \text{kg}\cdot\text{m}^{-2}$) vegetated areas, whereas in densely vegetated ($\text{VWC} > 4.35\ \text{kg}\cdot\text{m}^{-2}$) areas, nighttime products showed better accuracy than daytime products across all bands. This finding may be explained by the fact that VWC attenuates SM emissions and replaces them with its own emissions [63]. In sparsely vegetated areas, which are mostly under water stress conditions and comprised of isohydric species (e.g., open shrubs), the plants close its stomata during the daytime, thus keeping leaf water potential relatively steady during daytime [113], [114], [115]. This can lead to higher temperatures in the surrounding air, as there is less cooling effect from vegetation. In general, thermal equilibrium condition is not required for all SM retrieval algorithms, but it can significantly impact the accuracy of SM retrievals from passive microwave system [116]. Specifically, the retrieval of SM from AMSR2 using the LPRM algorithm relies on the brightness temperature, which makes thermal equilibrium a critical factor for precise results [117]. This is because if the observed surface is not in thermal equilibrium, the amount of microwave radiation measured by the system is not accurately representing the surface properties [16]. The attainment of thermal equilibrium depends on environmental conditions at the time of satellite overpass. For instance, areas can be energy-limited or water limited at any time during the day, which affects the relationship between evapotranspiration and land surface temperature, making it a nonlinear phenomenon [118]. Therefore, thermal equilibrium can be met during the daytime or at night, regardless of vegetation conditions. In addition, the presence of water bodies or winds can also influence thermal equilibrium in these areas [119]; thus, further studies are needed to comprehensively understand this phenomenon. In moderately vegetated areas, during the daytime, vegetation loses water through evapotranspiration and becomes transparent to microwave wavelengths, allowing precise retrieval of SM [51], [59], [111]. Conversely, in densely vegetated areas, nighttime products are more suitable than daytime products due to constant temperature conditions [52], [54], [120]. The LPRM algorithm assumes that vegetation temperature is equivalent to bare land surface temperature, which is indeed true over densely vegetated cover during the nighttime; however, during the daytime, increased evapotranspiration causes cooling which creates a significant difference between canopy and soil surface temperatures.

In this study, we concluded that stomatal closure due to water stress conditions in sparsely vegetated areas and evapotranspiration-induced vegetation transparency in moderately vegetated areas caused accurate retrieval of AMSR2 SM during the daytime. Whereas over densely vegetated areas, the assumption of the LPRM algorithm holds true during the nighttime due to the absence of solar radiations.

D. Error Characteristics of AMSR2 SM Based on Land Cover

The performance of AMSR2 SM was evaluated for seven different land cover types to identify points of accuracy and error in relation to underlying conditions. Fig. 6 shows violin plots of the fMSE for all six AMSR2 SM products in different land cover classes. In general, all six AMSR2 SM products showed reasonable accuracy across all seven land cover classes, with a median fMSE of less than 0.5 (i.e., the SM signal was stronger than its noise).

Specifically, in open shrublands, daytime overpass products outclassed nighttime overpass products. Fig. 6(a) shows a lower median fMSE for daytime overpass products when compared to nighttime overpass products. The same can be inferred from the shape of the kernel distribution: the section representing nighttime overpass products becomes wider as it tends toward low fMSE, indicating that low fMSE values are quite probable. In woody savannas [Fig. 6(b)], savannas [Fig. 6(c)], and grasslands [Fig. 6(d)], all AMSR2 products showed similar performance with a median fMSE of less than 0.4 and a kernel density distribution tending toward lower fMSE values. In croplands [Fig. 6(e)] and cropland/natural vegetation mosaics [Fig. 6(f)], all six products showed similar performance in terms of fMSE, and no significant differences were observed across different bands or overpass time products. However, in barren lands [Fig. 6(g)], X-band products showed slightly better performance than C-band products, and nighttime overpass products performed better than daytime overpass products. One plausible reason for this difference is that barren lands ($\text{VWC} < 0.1\ \text{kg}\cdot\text{m}^{-2}$) are characterized by extremely dry soils where high systematic retrieval errors make microwave-based SM retrievals problematic [93]. These systematic errors are mainly caused by uneven sensing depth,

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effective temperature, surface heterogeneity, and microwave emissions from subsurface [41], [75], [121], [122]. Moreover, arid regions have relatively little SM fluctuation, making it difficult to separate SM signals from the background noise of the instrument. As far as overpass time retrievals are concerned, the uneven distribution of soil temperature profiles across barren lands may lead SM retrieval issues during the daytime, while the isothermal conditions of these lands lead to better nighttime performance in the absence of solar radiations. In general, all three bands of AMSR2 produced large systematic errors in SM retrieval from dry regions [93]. These systematic errors are caused by the dryness of the soil, which makes estimating the thickness and effective temperature of the emitting layer problematic. Moreover, deep penetration of low-frequency (C-band) microwaves into dry soils reveals subsurface heterogeneity, again leading to uncertainty in SM retrieval [41], [74], [123]. In terms of different overpass products, the nighttime products are slightly better than the daytime products. This might be attributed to the fact that isothermal conditions and minimum Faraday rotations predominate in barren lands, causing fewer errors in nighttime overpasses than in daytime overpasses.

E. Selective Use of AMSR2 SM Products in Different Vegetation Conditions

As described in Sections IV-B and IV-C, AMSR2 products demonstrate unique performance and error characteristics across different vegetation and land cover classes. Specifically, this study found that C-band products outperformed X-band products in moderately and highly vegetated areas; and daytime products performed better than nighttime products in barren lands and sparsely vegetated areas. As a result, we believe that simultaneous use of products with complementary abilities may enhance the reliability of AMSR2 SM data. In this section, we evaluated the performance of selective combinations of AMSR2 products based on minimum fMSE. Fig. 7(a) shows a spatial distribution map of the fMSE from a combination of six AMSR2 products based on the lowest fMSE of the products. We can clearly see that selective use of AMSR2 SM products may dramatically reduce uncertainties on a global scale. Fig. 7(b) shows the spatial distribution of six AMSR2 SM products (based on frequency bands and overpass time) having minimum fMSE after selective use. Fig. 7(c) shows that the $F_{\mathrm{fMSE}}(0.5)$ values for all vegetation conditions are close to 1. For sparsely vegetated conditions, the value is 0.95; for moderately vegetated conditions, the value is 0.97; and for densely vegetated conditions, the value is 0.94. However, when the products were used individually, all fMSE values fell within the range of 0.6–0.8. The probability density functions (PDFs) and fMSE for all conditions show very similar distributions. Taken together, these findings show that selective use of all available AMSR2 SM products over different vegetation conditions provides a very high probability (95%) of producing SM data having stronger signal than noise. Thus, selective use of AMSR2 SM products shows great potential for remedying the shortcomings of individual products in challenging surface regions; however, this method does result in a 70% pixel loss of important data points.

V. Conclusion

The major goal of this study was to assess the global-scale error characteristic of AMSR2 SM products. AMSR2 is unique in providing SM in three different bands (C1, C2, and X) at the same overpass times; and being a sun-synchronous satellite, it can derive SM products from both daytime and nighttime overpasses. The products of different bands and acquisition times are influenced by contrasting underlying surface conditions, diurnal variability in land surfaces, and systematic SM retrievals errors; this study compared the global-scale error characteristics of AMSR2 SM products in different vegetation conditions and land cover classes using TCA. This comparison reveals the benefits and limitations of AMSR2 SM products as they relate to various vegetation conditions and land cover classes. The microwave remote sensing community will benefit from understanding the accuracy of AMSR2 products and using appropriate products for application in specific areas based on underlying vegetation gradients.

The TCA results showed that low-frequency (C-band) products were best at predicting SM over areas of high vegetation (excepting dense forests), with F_{fMSE} values of 0.6 improving to a range from 0.7 to 0.8. In sparsely and moderately vegetated areas, all products showed accurate F_{fMSE} values, improving in the range of 0.7–0.8. With regard to overpass times, daytime product performed better than nighttime in sparsely vegetated areas, specifically. However, in areas of high vegetation, C-band is slightly better than X-band. With regard to overpass time, and opposite to sparsely vegetated area is true, that is nighttime products outperformed daytime overpass products and the F_{fMSE} values of 0.5 improved to a range of 0.7.

By selectively using AMSR2 products in accordance with the results of TCA, we were able to obtain SM datasets of higher SNR. This encouraging result indicates that high-quality, global-scale SM datasets can be obtained over different vegetation and land cover conditions. While the selective use of AMSR2 SM products caused errors in many areas with impaired spatial coverage, overall, the selective use of different products can potentially contribute to improvements in the application of AMSR2 SM products in fields such as agriculture water management, water resource planning, numerical weather prediction, and understanding land surface–atmosphere interactions.

The Copernicus Imaging Microwave Radiometer (CIMR) mission, which will use several channels, including L-, C-, X-, Ka-, and Ku-bands, for the simultaneous retrieval of SM and vegetation properties, is currently under development. Our research will be extended once the SM data from CIMR becomes available in the future.

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available. Data supporting the conclusions of this study are properly cited and publicly available. Advanced Microwave Scanning Radiometer 2 (AMSR2) datasets are available at https://disc.gsfc.nasa.gov/datasets/LPRM_AMSR2_SOILM2_001/summary. Soil Moisture Active and Passive (SMAP) L3 vegetation water content (VWC) and International Geosphere Biosphere Program (IGBP) land classification datasets are available at https://nsidc.org/data/SPL2SMP/versions/7. Advanced Scatterometer (ASCAT) datasets are available at https://land.copernicus.vgt.vito.be/. Global Land Data Assimilation System (GLDAS) data are available at https://disc.sci.gsfc.nasa.gov/.

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